

SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1518

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 50

Question Count: 4

Code of Conduct

I K.Vignesh (192472154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Vignesh

Assessment Tasks Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

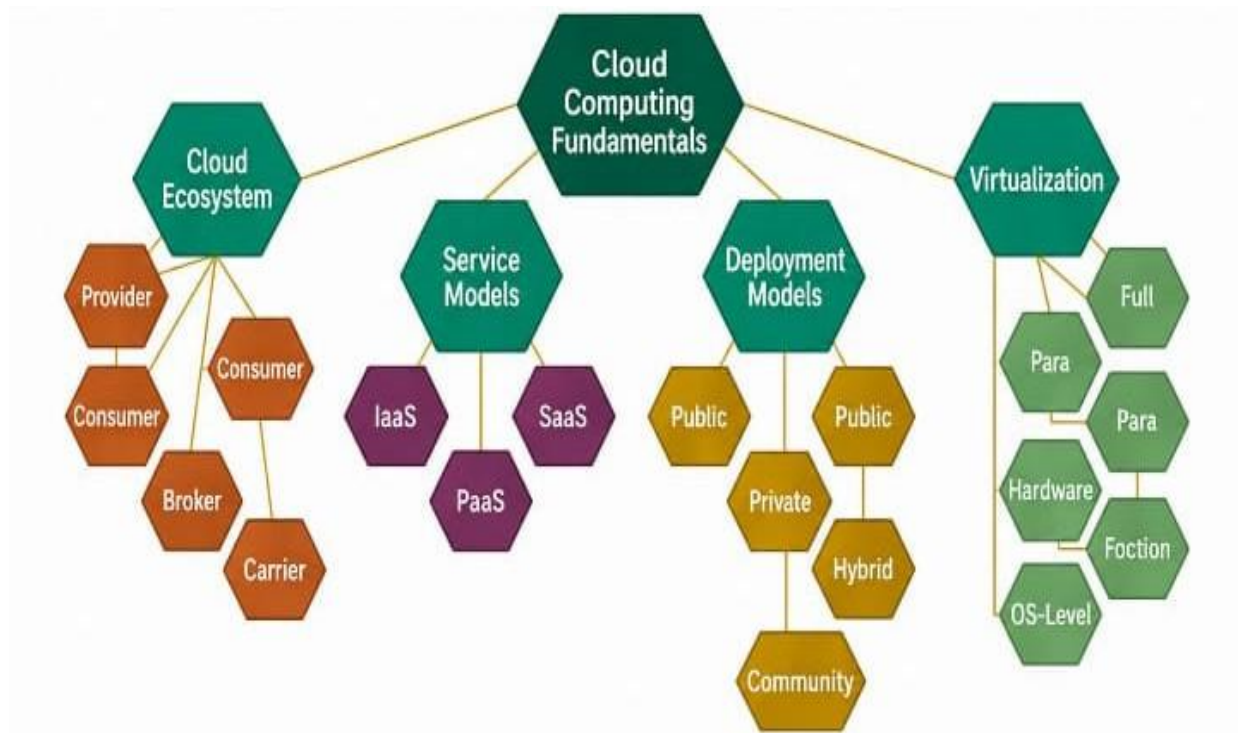
Concept Mapping Rubric

Criteria	Weightage	Level 4 -	Level 3 -	Level 2 -	Level 1 -
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		Exemplary	Proficient	Developing	Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
		accurate and meaningful			
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

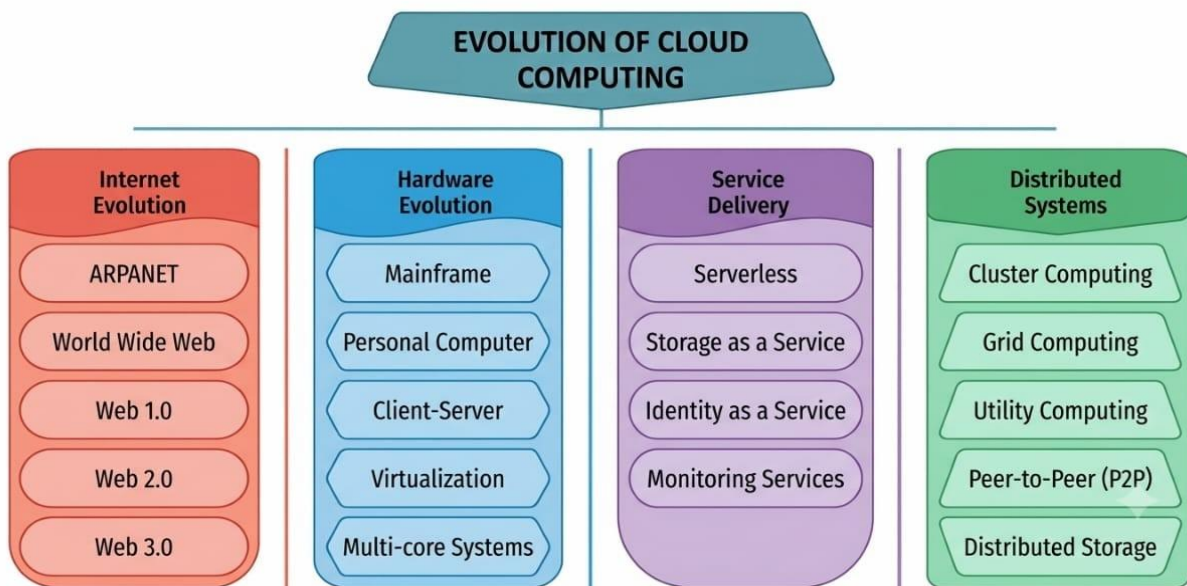
Concept Map 1

Cloud Computing Fundamentals. Include definition, characteristics, benefits, and Limitations



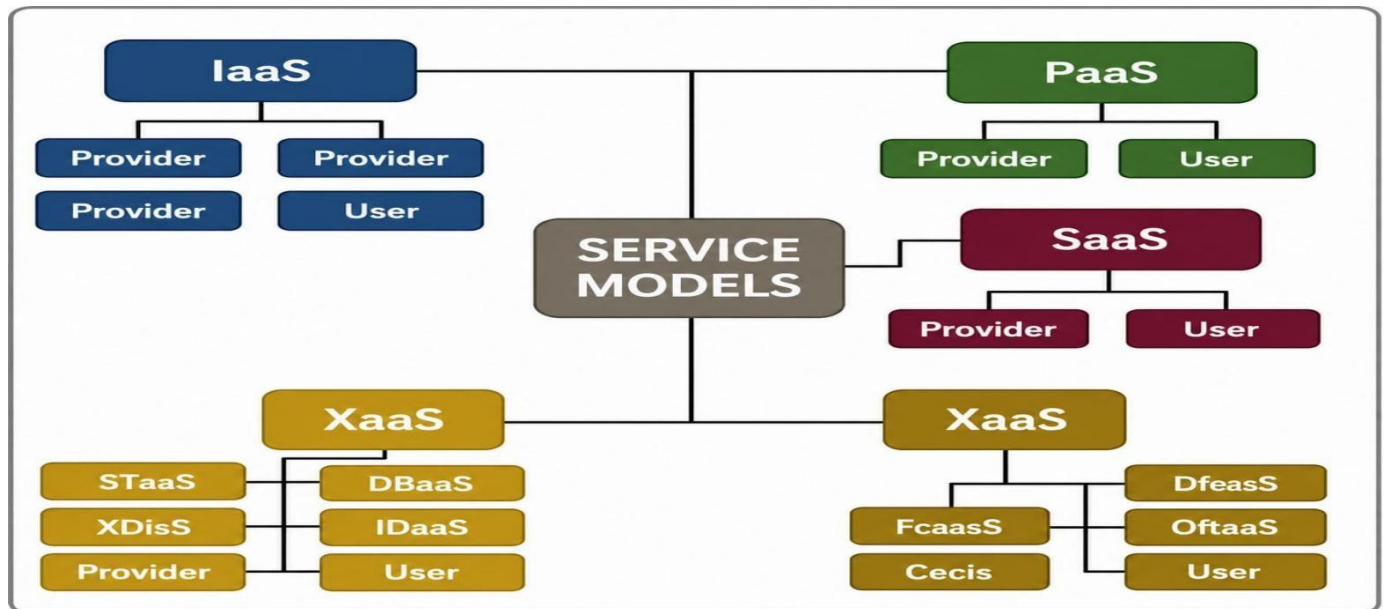
Concept Map 2

Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



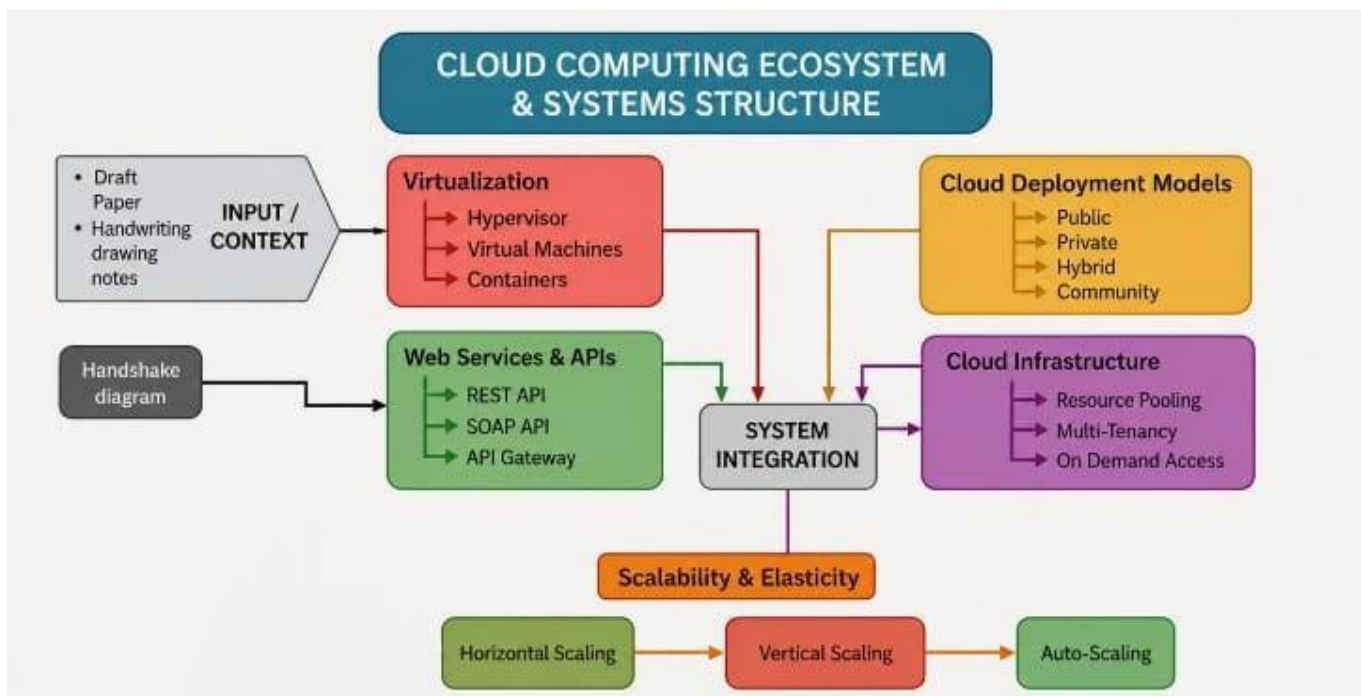
Concept Map 3

Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Concept Map 4

Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map clearly represents cloud computing by organizing concepts in a logical flow from input, virtualization, and web services to cloud infrastructure, system integration, and scalability. The color coding and arrows make the relationships easy to understand.

Effectiveness

- Clearly represents the key principles of cloud computing.
- Organizes concepts from general to specific.
- Makes complex cloud concepts easier to understand.
- Shows the relationship between virtualization, web services, deployment models, and scalability.

Design Decisions

- Used Cloud Computing as the central concept.
- Connected related concepts using branches and arrows.
- Placed each concept in a separate box for clarity.
- Followed a hierarchical structure to improve readability and logical flow.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, elasticity, and resource management are essential features of cloud computing architecture because they ensure that computing resources are efficiently utilized according to user demand. Scalability allows the system to increase or decrease computing resources as workload changes, while elasticity enables these adjustments automatically and in real time. Resource management coordinates the allocation of CPU, memory, storage, and network resources to maximize efficiency and reduce operational costs.

Influence on the architecture:

- Ensures efficient allocation of computing resources.
- Supports high availability during peak workloads.
- Reduces infrastructure costs through optimized resource usage.
- Improves application performance and response time.
- Enables automatic scaling based on user demand.
- Prevents resource wastage during low-demand periods.

Challenges if these capabilities were absent:

- Poor system performance during high traffic.
- Increased downtime and service interruptions.
- Higher operational costs due to inefficient resource utilization.
- Difficulty in handling sudden increases in user requests.
- Reduced reliability and lower customer satisfaction.
- Greater administrative effort for manual resource allocation.

Analysis: These capabilities form the foundation of modern cloud computing because they provide flexibility, efficiency, reliability, and cost-effectiveness.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

Contribution of each component:

- REST API: Provides lightweight, fast, and scalable communication using HTTP.
- SOAP API: Ensures secure and reliable message exchange for enterprise applications.
- API Gateway: Manages authentication, authorization, traffic routing, monitoring, and load balancing.
- Cloud Deployment Models: Offer flexible deployment options based on organizational requirements.
- Cloud Infrastructure: Provides computing power, storage, networking, and resource pooling.

Overall benefits:

- Faster communication between applications.
- Improved interoperability across multiple platforms.
- Enhanced security through centralized access control.
- Better scalability and system performance.
- Simplified integration of cloud services.
- Efficient management of cloud resources.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

Virtualization

- Enables resource sharing.
- Creates virtual machines.
- Improves hardware utilization.

Relationships between components:

Virtualization optimizes hardware utilization.
Web Services and APIs enable seamless communication.
Cloud Infrastructure provides shared computing resources.
Deployment Models determine how services are delivered.
System Integration connects all cloud components.
Scalability and Elasticity continuously optimize resource allocation.

Benefits of these interactions:

Increased reliability through resource redundancy.
Improved application performance with dynamic resource allocation.
Faster response times for users.
Enhanced user experience through continuous service availability.
Better fault tolerance and disaster recovery.
Efficient collaboration among distributed cloud services.

5. **Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?**

Learning outcomes:

Improved understanding of cloud architecture.
Better visualization of relationships between components.
Enhanced analytical and organizational skills.
Stronger understanding of scalability and virtualization.
Greater appreciation of cloud service integration.
Increased confidence in explaining cloud computing concepts.

Improvements for future redesign:

Include security components such as Identity and Access Management (IAM), encryption, and firewalls.
Add monitoring and logging services.
Include load balancing and disaster recovery mechanisms.
Represent data flow more clearly with additional directional arrows.
Add examples of popular cloud providers such as AWS, Microsoft Azure, and Google Cloud.

